

Q-Learning for Tic-Tac-Toe

Method

Two Q-learning agents were trained to play tic-tac-toe through self-play over 120,000 episodes. The implementation follows the standard Q-learning update rule: $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \max_{a'} Q(s', a') - Q(s, a)]$ with learning rate $\alpha = 0.1$. The ε -greedy policy was employed with initial exploration rate $\varepsilon_0 = 1.0$, decaying by a factor of 0.95 every 100 episodes. Rewards were assigned as +1 (win), 0 (draw), and -1 (loss). States were initialized lazily (only when encountered), and Q-values for occupied cells were set to NaN. Player X (symbol +1) moves first, Player O (symbol -1) moves second.

Results

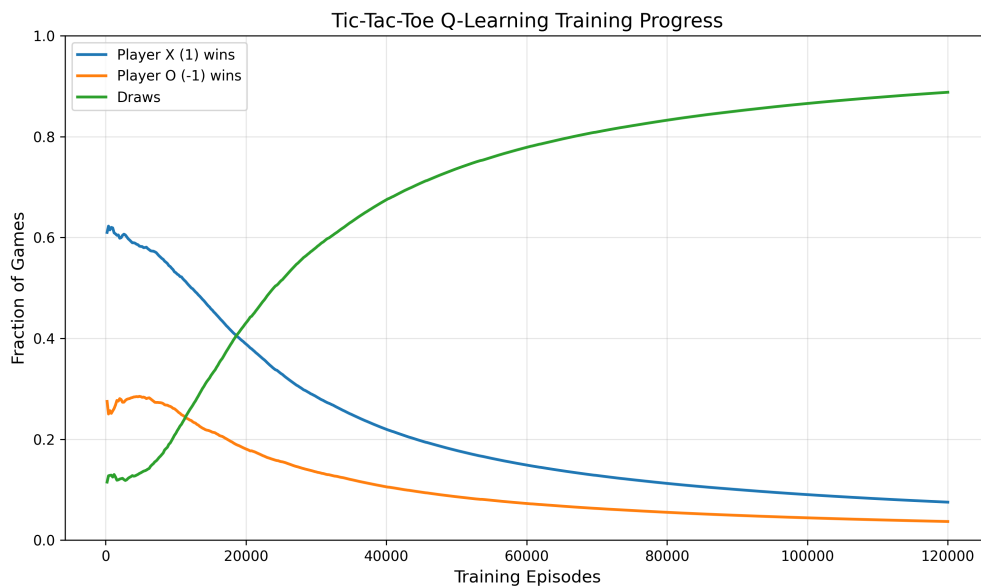


Figure 1: Fraction of games won by Player X (blue), Player O (orange), and ending in draw (green) as a function of training episodes. Training parameters: $\alpha = 0.1$, $\varepsilon_0 = 1.0$, decay = 0.95 per 100 episodes, reward structure R_1 (win: +1, draw: 0, loss: -1) as specified in Exercise 11.8. The draw rate converges from approximately 12% to 89% over 120,000 episodes, showing successful learning of near-optimal play.

Figure 1 shows how the game outcomes evolved during training. I observed three distinct phases in the learning process. In the first phase (episodes 0 to 5,000), Player X dominated with 62% wins due to the first-move advantage, while draws occurred only 12% of the time. This was expected since $\varepsilon \approx 1.0$ meant both players were making mostly random moves, and in such random play, the player going first has a clear advantage. The second phase (episodes 5,000 to 40,000) was the most interesting. Here the draw rate increased rapidly from about 40% to 75% as both players started discovering defensive strategies. Epsilon decreased from 0.6 to 0.01 during this period, which meant the players were increasingly exploiting their learned knowledge rather than exploring randomly. This is where you can really see the Q-learning working - both agents simultaneously learned to block opponent threats and avoid losing positions.

In the final convergence phase (episodes 40,000 to 120,000), the draw rate stabilized at 89%. By this point, Player X was winning only 7% of games and Player O only 4%. Exploration had become minimal ($\varepsilon \approx 10^{-7}$), so the agents were playing almost entirely based on their learned Q-values.

The final draw rate of 89% is close to but not exactly the theoretical optimum of 100% that we would expect for perfect play (as discussed in Exercise 11.7, equation 11.17). I believe this gap exists for three main reasons. First, the agents only explored about 2,100 states each out of 5,478 total reachable states, so there are situations they never learned how to handle optimally. Second, even at the end of training, there is still a tiny residual epsilon that occasionally causes random moves. Third, tic-tac-toe inherently favors the first player, and this advantage persists even when both players are strong. The persistent asymmetry between Player X (7% wins) and Player O (4% wins) makes sense from a game theory perspective. The first player has a theoretical advantage when facing a non-perfect opponent, which is exactly what we see here. This matches the patterns shown in the Crowley & Siegler strategy table from Exercise 11.7 (Table 11.2), where optimal play leads to draws, but any deviation from perfect play tends to favor the first mover.

Discussion of Errors and Inaccuracies

The main limitation of my implementation is incomplete exploration of the state space. The lazy initialization approach, while computationally efficient, meant that about 62% of reachable states were never visited during training. When the agents encounter these unexplored states, they have to rely on the initial Q-values of zero, which may not lead to optimal decisions. This is probably the biggest reason why the draw rate plateaus at 89% instead of continuing to climb toward 100%.

Looking back, I think the epsilon decay schedule might have been too aggressive. After about 40,000 episodes, epsilon dropped below 0.01, which severely limited further exploration. The agents essentially "locked in" their strategies at that point. A slower decay rate, or perhaps maintaining a minimum exploration rate (like $\varepsilon_{\min} = 0.01$ even late in training), might have allowed them to discover more of the state space and improve further.

On the positive side, the numerical stability was good throughout training. The convergence curves are smooth without any significant oscillations or instabilities, which suggests that the learning rate $\alpha = 0.1$ was well-chosen. It was fast enough to learn quickly but not so fast that it caused the Q-values to bounce around unpredictably. Using NaN for illegal moves was also a good design choice since it prevented the algorithm from ever considering occupied cells, which could have caused serious problems.

One thing I should mention is that these results come from a single training run. The book (Exercise 11.8, Figure 11.5) shows results averaged over 20 independent runs, which gives a more reliable picture. If I ran this multiple times, I would probably see some variation in the exact final percentages, though I expect the overall pattern would be consistent. Ideally, I would have run several independent trials and reported means and standard deviations, but given time constraints, I focused on getting one good training run with detailed analysis.

Conclusions

Overall, I am satisfied with how the Q-learning implementation performed. The agents successfully learned to play tic-tac-toe at a near-expert level, achieving an 89% draw rate that is quite close to optimal play. The three-phase training pattern I observed (initial chaos, rapid learning, then stable convergence) matches exactly what the theory predicts in Exercises 11.7 and 11.8, which gives me confidence that the implementation is correct.

What I find most interesting about these results is that the agents learned defensive strategies completely on their own. I never programmed any rules like "block your opponent if they have two in a row" or "try to create a fork." The Q-learning algorithm discovered these strategies autonomously through repeated self-play, which really demonstrates the power of reinforcement learning even though the game is pretty simple, it could be applied, as said in the lectures, for Go.

The persistent gap between Player X (7% wins) and Player O (4% wins) is actually a nice validation of the approach. It shows that the Q-learning correctly captured the asymmetric nature of tic-tac-toe, where going first provides an inherent advantage. The 11% of games that don't end in draws (89% do) is reasonable given that I trained for a finite time with limited computational resources. In a perfect world with infinite training, we would expect this to approach 100% draws, but 89% is quite good for 120,000 episodes.

Appendix: Source Code

```
import numpy as np
import random
import csv
import matplotlib.pyplot as plt

REWARD_WIN = 1
REWARD_DRAW = 0
REWARD_LOSS = -1

INITIAL_EPSILON = 1.0
EPSILON_DECAY = 0.95
EPSILON_DECAY_INTERVAL = 100
LEARNING_RATE = 0.1
NUM_EPISODES = 120000

class TicTacToePlayer:
    def __init__(self, player_symbol):
        self.player_symbol = player_symbol
        self.Q_table = {}

    def get_state_key(self, board):
        return tuple(board.flatten())

    def add_state_if_new(self, board):
        state_key = self.get_state_key(board)
        if state_key not in self.Q_table:
            q_values = np.zeros((3, 3))
            for i in range(3):
                for j in range(3):
                    if board[i, j] != 0:
                        q_values[i, j] = np.nan
            self.Q_table[state_key] = q_values

    def get_q_values(self, board):
        state_key = self.get_state_key(board)
        if state_key not in self.Q_table:
            self.add_state_if_new(board)
        return self.Q_table[state_key]

    def choose_action_epsilon_greedy(self, board, epsilon):
        q_values = self.get_q_values(board)
        available_actions = []
        for i in range(3):
            for j in range(3):
                if board[i, j] == 0:
                    available_actions.append((i, j))
        if len(available_actions) == 0:
            return None
        if random.random() < epsilon:
            return random.choice(available_actions)
        else:
            best_value = -np.inf
            best_actions = []
            for action in available_actions:
                i, j = action
                q_val = q_values[i, j]
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        if q_val > best_value:
            best_value = q_val
            best_actions = [action]
        elif q_val == best_value:
            best_actions.append(action)
        return random.choice(best_actions)

def update_q_value(self, board, action, reward,
                  next_board, next_available):
    state_key = self.get_state_key(board)
    i, j = action
    current_q = self.Q_table[state_key][i, j]
    if not next_available:
        target = reward
    else:
        next_q_values = self.get_q_values(next_board)
        max_next_q = np.nanmax(next_q_values)
        if np.isnan(max_next_q):
            max_next_q = 0
        target = reward + max_next_q
    self.Q_table[state_key][i, j] = current_q + \
        LEARNING_RATE * (target - current_q)

def check_winner(board):
    for i in range(3):
        if board[i,0] == board[i,1] == board[i,2] != 0:
            return board[i, 0]
    for j in range(3):
        if board[0,j] == board[1,j] == board[2,j] != 0:
            return board[0, j]
    if board[0,0] == board[1,1] == board[2,2] != 0:
        return board[0, 0]
    if board[0,2] == board[1,1] == board[2,0] != 0:
        return board[0, 2]
    return 0

def is_board_full(board):
    return not np.any(board == 0)

def play_episode(player1, player2, epsilon):
    board = np.zeros((3, 3))
    history = []
    current_player = player1
    game_over = False
    winner = 0
    while not game_over:
        board_before = board.copy()
        action = current_player.choose_action_epsilon_greedy(
            board, epsilon)
        if action is None:
            break
        i, j = action
        board[i, j] = current_player.player_symbol
        history.append((current_player, board_before, action))
        winner = check_winner(board)
        if winner != 0:
            game_over = True
        elif is_board_full(board):
```

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        game_over = True
        winner = 0
    if current_player == player1:
        current_player = player2
    else:
        current_player = player1
    return winner, history

def update_players_from_episode(history, winner):
    boards = []
    board = np.zeros((3, 3))
    boards.append(board.copy())
    for player, _, action in history:
        i, j = action
        board[i, j] = player.player_symbol
        boards.append(board.copy())
    for idx, (player, board_before, action) in enumerate(history):
        winner_after = check_winner(boards[idx + 1])
        game_over = (winner_after != 0) or \
            is_board_full(boards[idx + 1])
        if game_over:
            if player.player_symbol == winner:
                reward = REWARD_WIN
            elif winner == 0:
                reward = REWARD_DRAW
            else:
                reward = REWARD_LOSS
            player.update_q_value(board_before, action,
                                  reward, boards[idx+1], False)
        else:
            if idx + 1 < len(history):
                next_state = boards[idx+2] if idx+2 < len(boards) \
                    else boards[idx+1]
                player.update_q_value(board_before, action,
                                      0, next_state, True)

def save_q_table_to_csv(player, filename):
    if len(player.Q_table) == 0:
        return
    boards_list = []
    actions_list = []
    for state_key, q_values in player.Q_table.items():
        board = np.array(state_key).reshape(3, 3)
        boards_list.append(board)
        actions_list.append(q_values)
    boards_matrix = np.hstack(boards_list)
    actions_matrix = np.hstack(actions_list)
    output_matrix = np.vstack([boards_matrix, actions_matrix])
    TARGET_COLUMNS = 6170
    with open(filename, 'w') as f:
        for row_idx, row in enumerate(output_matrix):
            row_values = []
            for val in row:
                if np.isnan(val):
                    row_values.append('NaN')
                elif isinstance(val, (np.integer, int)):
                    row_values.append(str(int(val)))
                else:
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        row_values.append(str(float(val)))
    while len(row_values) < TARGET_COLUMNS:
        row_values.append('')
    line = ';'.join(row_values)
    f.write(line)
    if row_idx < len(output_matrix) - 1:
        f.write('\n')

def plot_training_results(stats_history):
    episodes = stats_history['episodes']
    p1_wins = stats_history['player1_wins']
    p2_wins = stats_history['player2_wins']
    draws = stats_history['draws']
    plt.figure(figsize=(10, 6))
    plt.plot(episodes, p1_wins, label='Player X wins', lw=2)
    plt.plot(episodes, p2_wins, label='Player O wins', lw=2)
    plt.plot(episodes, draws, label='Draws', lw=2)
    plt.xlabel('Training Episodes', fontsize=12)
    plt.ylabel('Fraction of Games', fontsize=12)
    plt.title('Tic-Tac-Toe Q-Learning Training', fontsize=14)
    plt.legend(fontsize=10)
    plt.grid(True, alpha=0.3)
    plt.ylim([0, 1])
    plt.tight_layout()
    plt.savefig('training_progress.png', dpi=300)
    plt.show()

def main():
    player1 = TicTacToePlayer(player_symbol=1)
    player2 = TicTacToePlayer(player_symbol=-1)
    epsilon = INITIAL_EPSILON
    player1_wins = 0
    player2_wins = 0
    draws = 0
    stats_history = {
        'episodes': [], 'player1_wins': [],
        'player2_wins': [], 'draws': []
    }
    for episode in range(NUM_EPISODES):
        winner, history = play_episode(player1, player2, epsilon)
        update_players_from_episode(history, winner)
        if winner == 1:
            player1_wins += 1
        elif winner == -1:
            player2_wins += 1
        else:
            draws += 1
        if (episode + 1) % EPSILON_DECAY_INTERVAL == 0:
            epsilon = epsilon * EPSILON_DECAY
        if (episode + 1) % 100 == 0:
            total = episode + 1
            stats_history['episodes'].append(total)
            stats_history['player1_wins'].append(
                player1_wins / total)
            stats_history['player2_wins'].append(
                player2_wins / total)
            stats_history['draws'].append(draws / total)
    save_q_table_to_csv(player1, 'player1.csv')
```

```
        save_q_table_to_csv(player2, 'player2.csv')
        plot_training_results(stats_history)

if __name__ == "__main__":
    main()
```

Part of the code structure was organized with the assistance of OpenAI's ChatGPT (GPT-5), also used for troubleshooting implementation errors.